



CAPSTONE PROJECT REGISTER

Class: _____ Duration time: from/.../.... To/... /.....

(*) Profession: < Artificial Intelligence > Specialty: CV ☒ NLP ☐ DS ☐ Other ☐

(*) Kinds of person make registers: Lecturer ☒ Students ☐

1. Register information for supervisor (if have)

	Full name	Phone	E-Mail	Title
Supervisor 1	Đặng Ngọc Minh Đức		ducnm2@fe.edu.vn	Assoc.Prof.

2. Register information for students (if have)

	Full name	Student code	Phone	E-mail	Role in Group
Student 1					Leader
Student 2					Member
Student 3					Member

3. Register content of the Capstone Project

(*) 3.1. Capstone Project name:

English: Federated Learning for Medical Image Classification

Vietnamese: Nghiên cứu học liên kết cho bài toán phân loại ảnh y tế

Abbreviation: FedMedAI

a. Context:

Medical image classification is one of the most important applications of AI in healthcare, supporting disease diagnosis and computer-aided diagnosis. However, medical data are naturally distributed across hospitals, and privacy regulations, data ownership, and institutional restrictions make centralized data collection impractical. Federated Learning (FL) enables multiple institutions to collaboratively train a global model without exchanging raw data — each participant trains locally and shares only model parameters. Open-source frameworks such as Flower now make FL practical to implement without large-scale infrastructure.

Research Gap:

- Non-IID heterogeneity: FL performance degrades sharply under non-IID data across clients — the norm in real hospitals — where simple FedAvg often underperforms centralized training.
- Communication overhead: FL requires many communication rounds; communication cost is a first-class constraint that is frequently under-reported.
- Residual privacy risk: even without sharing raw data, shared model updates can leak information, motivating privacy-enhancing techniques.
- Fairness across clients: heterogeneous clients can bias the global model against underrepresented institutions or rare disease classes.

Therefore, investigating lightweight, communication-efficient, and heterogeneity-robust FL strategies for medical imaging is an open, practically important research direction. This capstone targets a concrete, measurable technical contribution rather than a plain FL demo.

b. Objectives:

The objectives of this capstone project are:

- Study the principles of Federated Learning and privacy-preserving ML.
- Investigate distributed deep learning for healthcare.
- Implement a federated medical image classification system with Flower.
- Compare centralized learning and Federated Learning.
- Analyze the influence of non-IID data distribution (Dirichlet).
- Develop a prototype for collaborative model training.

Research objective (contribution):

Propose and experimentally validate at least one technical improvement over a FedAvg baseline (e.g., FedBN / FedProx / FedNova / SCAFFOLD, client sampling, or communication reduction) that measurably improves accuracy, robustness to non-IID data, or communication efficiency, supported by ablation and error analysis.

c. Proposed solution:

Dataset (scope-controlled):

Use MedMNIST v2 as the primary dataset (e.g., BloodMNIST, DermaMNIST, ChestMNIST, PathMNIST). Its lightweight, standardized 2D images make it ideal for federated simulation within a capstone timeframe. Large clinical datasets such as BraTS, ChestX-ray14, and ISIC are discouraged — their size and complexity exceed the scope of this capstone.

Framework (mandatory):

Use Flower (flwr) as the federated-learning framework. Students are not required to build an FL framework from scratch; the focus is on FL methodology and experimentation.

Baseline Requirement:

Establish two references before any improvement: a centralized-learning model and a FedAvg federated baseline. FedAvg is the mandatory FL baseline; improvements are measured against it.

Non-IID Protocol (standardized):

Simulate client heterogeneity with a Dirichlet distribution over labels, evaluated at multiple concentration levels:

- $\alpha = 0.1$ (highly non-IID).
- $\alpha = 0.3$ (moderately non-IID).
- $\alpha = 1.0$ (mild / near-IID).

Required Improvement (Novelty):

After the baselines are established, students must implement at least one of the following (or an equivalent contribution):

- FedBN — local batch normalization for feature-shift heterogeneity.
- FedProx — proximal-term optimization on the client objective (note: this is proximal optimization, not adaptive aggregation).
- FedNova — normalized averaging to correct objective inconsistency.
- SCAFFOLD — control variates to reduce client drift.

The improvement must be motivated by the Research Gap and evaluated against the FedAvg baseline under identical conditions.

Experimental Evaluation:

Compare Centralized vs. FedAvg vs. the proposed improvement across the three Dirichlet non-IID levels and different numbers of clients. Metrics:

- Accuracy, Precision, Recall, F1-score.
- Communication rounds and communication cost (bytes / rounds to target accuracy).
- Client fairness (variance / worst-case accuracy across clients).
- Convergence speed (rounds to target accuracy).
- Training time.

Ablation study (required):

- IID vs. non-IID ($\alpha = 0.1 / 0.3 / 1.0$).

- Number of clients (e.g., 3 / 5 / 10).
- FedAvg vs. the proposed method.
- With vs. without the proposed improvement.

Error analysis (required):

- Per-class performance, focusing on minority/rare disease classes.
- Effect of client heterogeneity and data imbalance.
- Failure-case and confusion-matrix discussion.

Evaluation protocol (for consistency across groups):

Report the Dirichlet alpha and partitioning scheme, random seed(s), hardware configuration, per-round training time, and total communication cost.

Prototype Development:

A demonstration system that simulates multiple FL clients, monitors training progress, displays performance, and visualizes communication rounds and per-client contributions.

d. Algorithm requirement

Training

- Dataset: MedMNIST v2 subsets, partitioned into clients via Dirichlet ($\alpha = 0.1 / 0.3 / 1.0$).
- Framework: Flower (flwr) with Python + PyTorch.
- Algorithms: centralized baseline + FedAvg (mandatory); plus at least one improvement (FedBN / FedProx / FedNova / SCAFFOLD).

Testing

- Metrics: Accuracy, Precision, Recall, F1; communication rounds & cost; client fairness; convergence speed; training time.
- Baseline comparison: centralized vs. FedAvg vs. the improvement across non-IID levels.
- Ablation and error analysis are mandatory.
- Evaluation protocol: report alpha, partitioning, seed, hardware, training time, communication cost.

Deployment

- Runtime environment: a single machine simulating multiple clients (or a small multi-node setup) with Python + PyTorch + Flower.

- Interface: a monitoring dashboard (client status, training curves, communication rounds, performance).

(*) 3.2. Main proposal content (including result and product)

a. **Theory and practice (document):**

Apply knowledge from AIL302m (AI), AIP391 (Machine Learning), DPL301m (Deep Learning), Distributed Systems, and Software Engineering. Literature review: Deep Learning for Medical Image Classification, Federated Learning, Privacy-Preserving ML (Differential Privacy, Secure Aggregation), Distributed Optimization, and Communication-Efficient Learning.

- Artificial Intelligence technologies:

Framework: PyTorch; Flower (flwr) for federated orchestration.

Algorithms: FedAvg, FedBN, FedProx, FedNova, SCAFFOLD.

Deployment: Python.

b. **Products:**

- A Federated Learning system (Flower-based) for medical image classification.
- Experimental reports comparing centralized and federated learning (with ablation studies).
- A prototype visualization / monitoring system.
- A complete capstone thesis; results encouraged toward undergraduate research competitions or publications.

Reproducibility package (required):

Students must provide:

- Source code.
- Trained model weights.
- Client-partitioning configuration (Dirichlet alpha, seeds).
- Configuration files.
- Inference script and README.
- Experiment log (metrics, seeds, settings per run).

Proposed Tasks (aligned with the AIP491 15-week schedule):

- **Task package 1 (Weeks 1–3):** Literature review, MedMNIST preparation, Flower environment setup (→ Report 1: Introduction; Report 3: Existing Systems).
- **Task package 2 (Weeks 4–6):** Centralized model implementation and baseline evaluation (→ Report 4: Methodology).

- **Task package 3 (Weeks 7–10):** FedAvg implementation, Dirichlet non-IID partitioning, and the proposed improvement (→ Report 4/5).
- **Task package 4 (Weeks 11–13):** Performance evaluation, ablation studies, error analysis, and comparative experiments across non-IID levels (→ Report 6 & 7: Results/Discussion).
- **Task package 5 (Weeks 14–15):** Prototype development, thesis writing, and final presentation (→ Final report and defense).

AI Ethics and Responsible AI:

This topic is inherently privacy-centric; consistent with AIP491 CLO7 and the AI Ethics and Data Privacy session:

- Patient privacy and data governance — why FL keeps raw data local, and residual leakage risk through shared updates.
- Privacy-enhancing techniques — Differential Privacy and Secure Aggregation (conceptual discussion; optional implementation).
- Fairness across institutions and populations — heterogeneous clients can bias the global model against underrepresented groups or rare conditions.
- Responsible deployment — the limits of automated diagnosis and the need for clinician oversight.

Expected Learning Outcomes:

Upon completion, students will be able to (mapped to AIP491 CLOs):

- Implement distributed / federated deep-learning systems with Flower (CLO3, CLO8).
- Reproduce centralized and FedAvg baselines, then propose and validate an improvement (CLO3, CLO8, CLO10).
- Conduct reproducible experiments with proper protocol, ablation, and error analysis (CLO2, CLO5, CLO9).
- Critically analyze non-IID effects, communication cost, and class imbalance (CLO9).
- Develop and deploy an interactive monitoring prototype (CLO3, CLO6).
- Apply privacy, fairness, and ethical standards to healthcare AI (CLO7).
- Communicate research findings effectively in a written thesis and oral defense (CLO5, CLO6).

References:

Foundations

[1] B. McMahan, et al. Communication-Efficient Learning of Deep Networks from Decentralized Data (FedAvg). AISTATS, 2017.

[2] T. Li, et al. Federated Optimization in Heterogeneous Networks (FedProx). MLSys, 2020.

Benchmark / Dataset

[3] J. Yang, et al. MedMNIST v2: A Large-Scale Lightweight Benchmark for 2D and 3D Biomedical Image Classification. Scientific Data, vol. 10, 2023.

Baseline / Representative Methods

[4] X. Li, et al. FedBN: Federated Learning on Non-IID Features via Local Batch Normalization. ICLR, 2021.

[5] S. P. Karimireddy, et al. SCAFFOLD: Stochastic Controlled Averaging for Federated Learning. ICML, 2020.

[6] J. Wang, et al. Tackling the Objective Inconsistency Problem in Heterogeneous Federated Optimization (FedNova). NeurIPS, 2020.

Frameworks & Privacy

[7] D. Beutel, et al. Flower: A Friendly Federated Learning Research Framework. arXiv:2007.14390, 2020.

[8] K. Bonawitz, et al. Practical Secure Aggregation for Privacy-Preserving Machine Learning. ACM CCS, 2017.

[9] M. Abadi, et al. Deep Learning with Differential Privacy. ACM CCS, 2016.

Survey

[10] P. Kairouz, et al. Advances and Open Problems in Federated Learning. Foundations and Trends in ML, 2021.

[11] D. C. Nguyen, et al. Federated Learning for Smart Healthcare: A Survey. ACM Computing Surveys, 2022.

4. Other comments (propose all relevant things if any)

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....., date / /

Supervisor (If have)
(Sign and full name)

On behalf of the Registers
(Sign and full name)